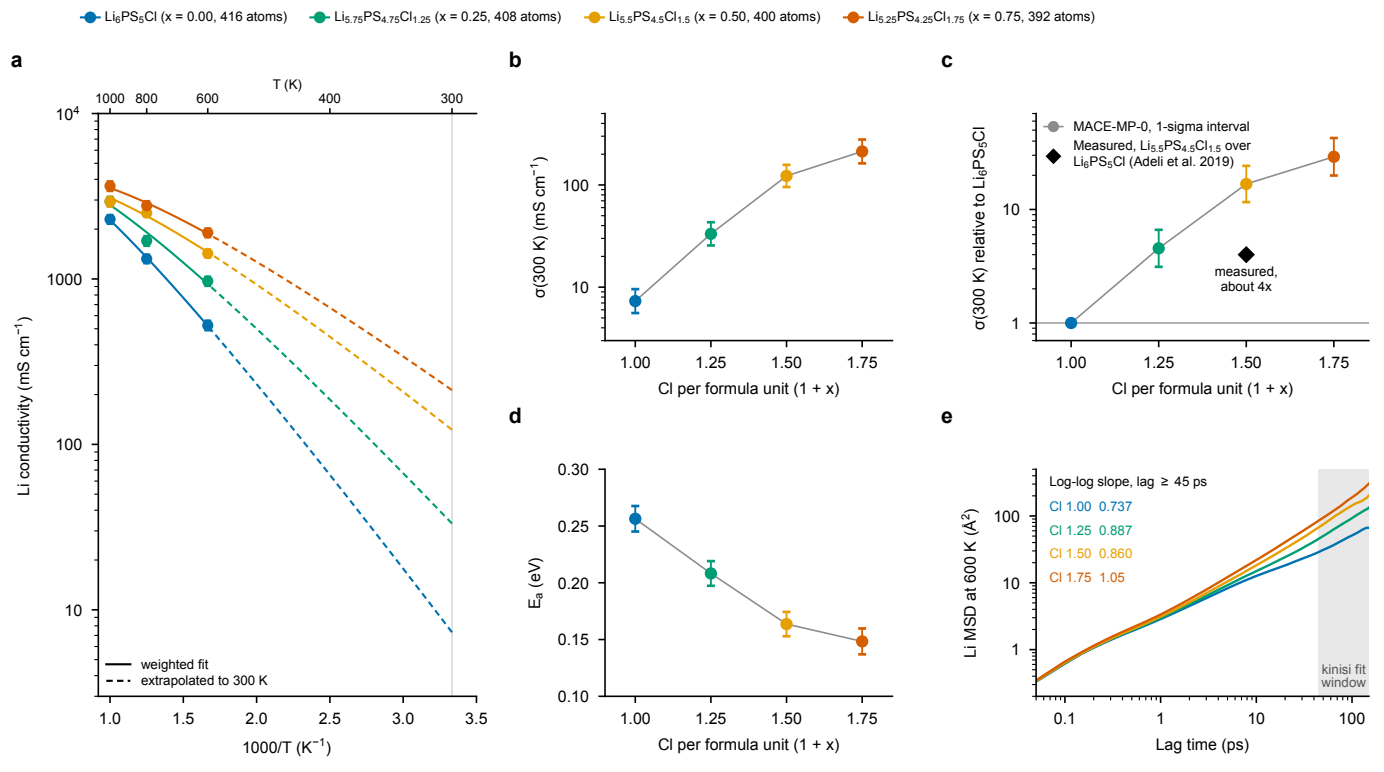

Li6PS5Cl (x = 0.00, 416 atoms)
Li5.75PS4.75Cl1.25 (x = 0.25, 408 atoms)
Li5.5PS4.5Cl1.5 (x = 0.50, 400 atoms)
Li5.25PS4.25Cl1.75 (x = 0.75, 392 atoms)
a
T (K)
1000 800 600 400 300
Li conductivity (mS cm−1)
1000/T (K−1)
weighted fit
extrapolated to 300 K
b
σ(300 K) (mS cm−1)
Cl per formula unit (1 + x)
c
MACE-MP-0, 1-sigma interval
Measured, Li5.5PS4.5Cl1.5 over Li6PS5Cl (Adeli et al. 2019)
measured, about 4x
σ(300 K) relative to Li6PS5Cl
Cl per formula unit (1 + x)
d
Ea (eV)
Cl per formula unit (1 + x)
e
Log-log slope, lag ≥ 45 ps
Cl 1.00 0.737
Cl 1.25 0.887
Cl 1.50 0.860
Cl 1.75 1.05
kinisi fit window
Li MSD at 600 K (Å2)
Lag time (ps)